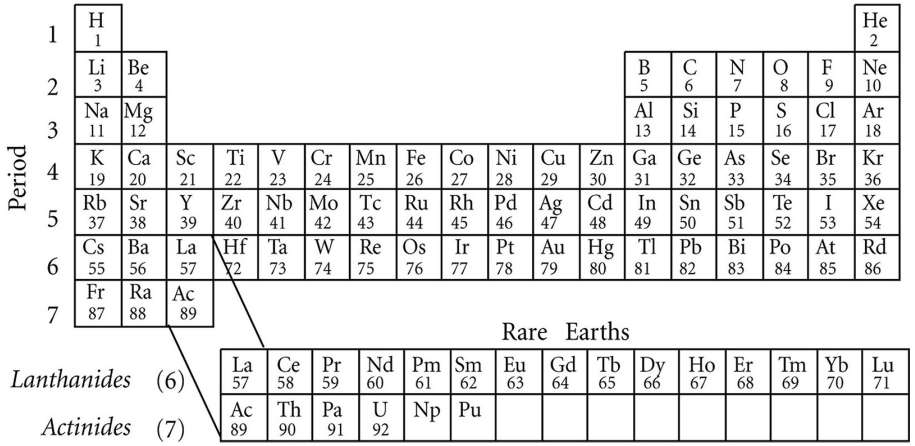
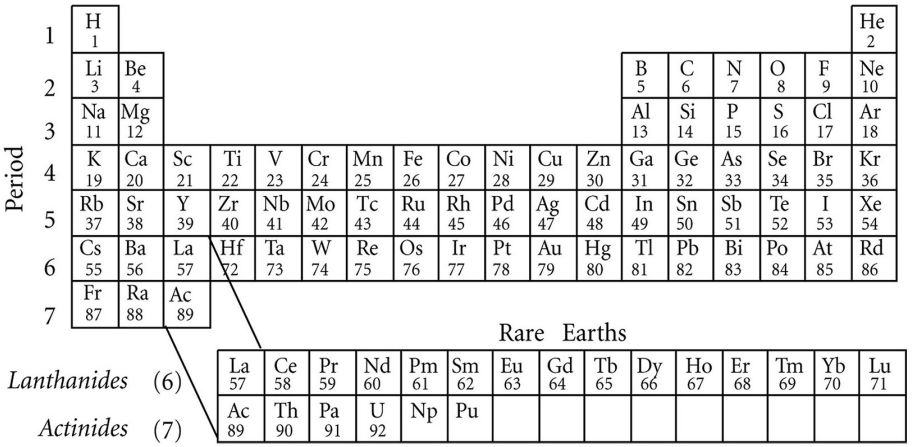
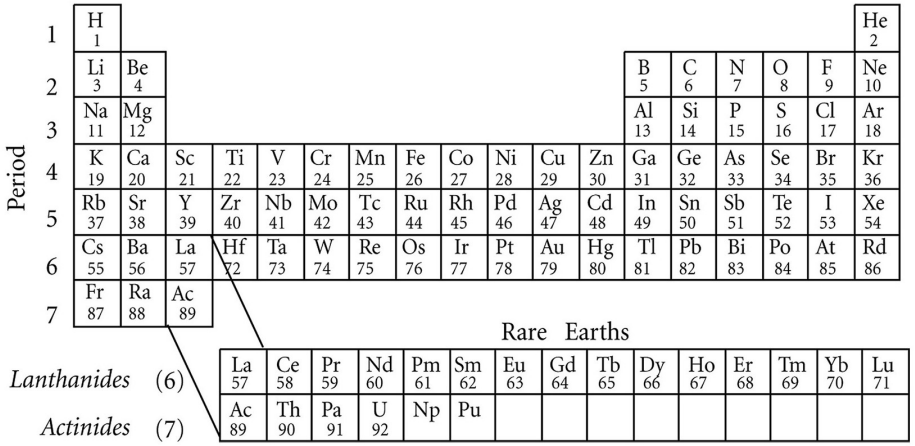
Chemistry of the Earth

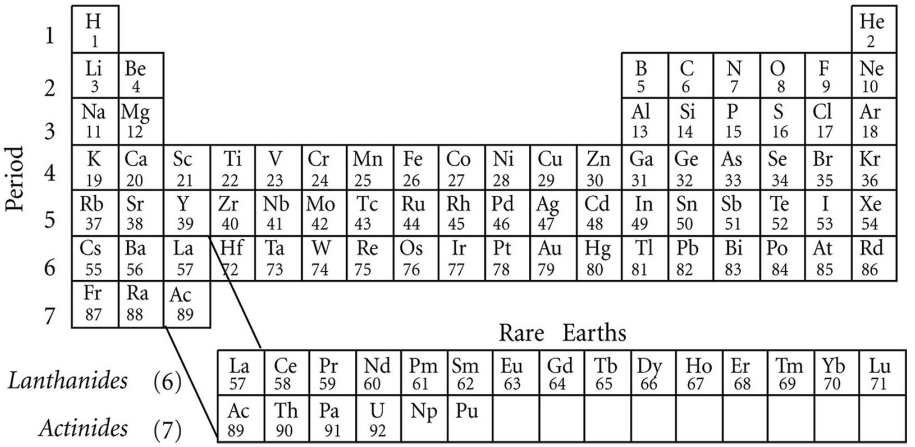
Overview of the Semester:

Short quiz on the Geochemist's Periodic Table

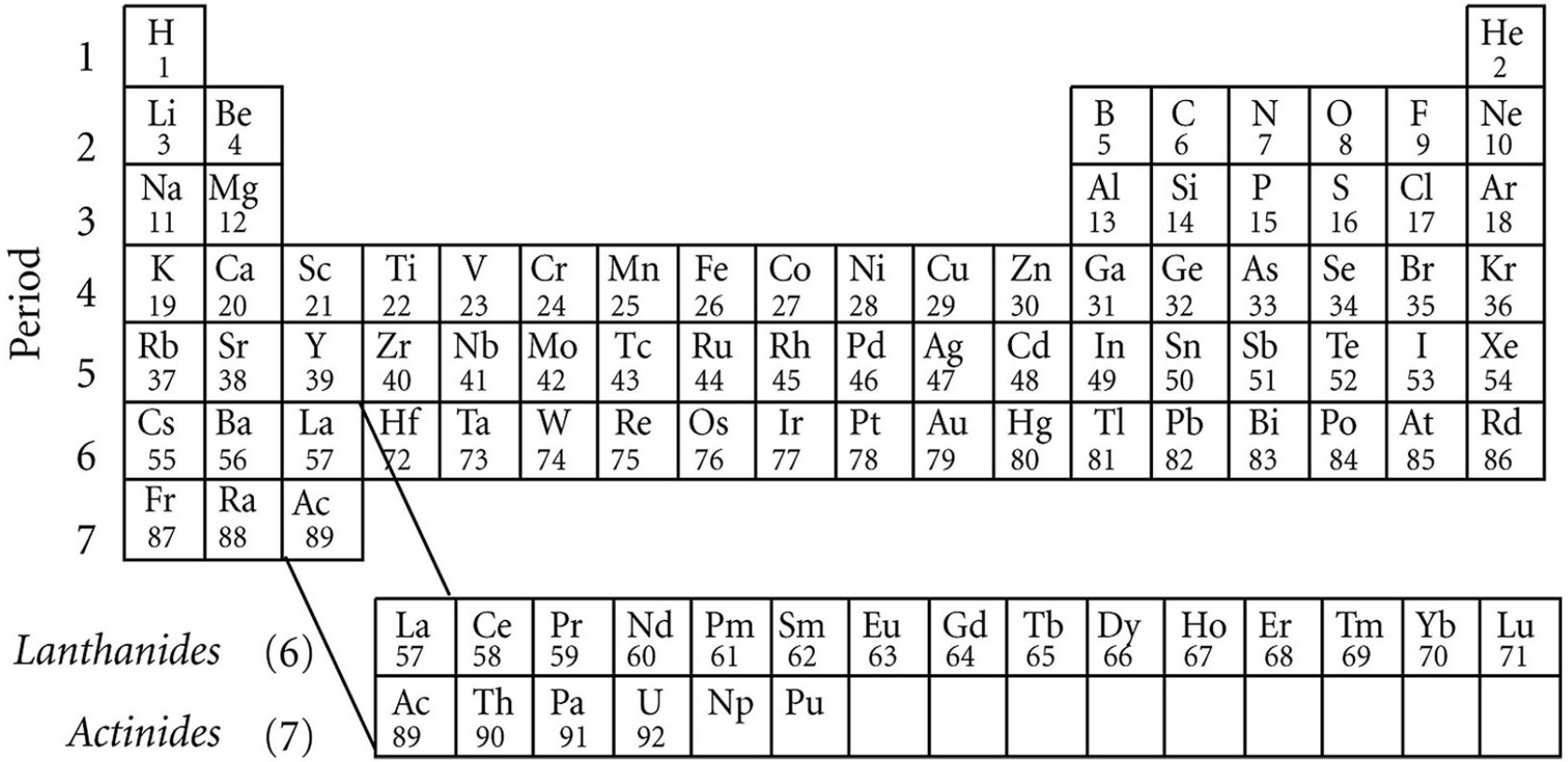

Period
1 H 1 He 2
2 Li 3 Be 4 B 5 C 6 N 7 O 8 F 9 Ne 10
3 Na 11 Mg 12 Al 13 Si 14 P 15 S 16 Cl 17 Ar 18
4 K 19 Ca 20 Sc 21 Ti 22 V 23 Cr 24 Mn 25 Fe 26 Co 27 Ni 28 Cu 29 Zn 30 Ga 31 Ge 32 As 33 Se 34 Br 35 Kr 36
5 Rb 37 Sr 38 Y 39 Zr 40 Nb 41 Mo 42 Tc 43 Ru 44 Rh 45 Pd 46 Ag 47 Cd 48 In 49 Sn 50 Sb 51 Te 52 I 53 Xe 54
6 Cs 55 Ba 56 La 57 Hf 72 Ta 73 W 74 Re 75 Os 76 Ir 77 Pt 78 Au 79 Hg 80 Tl 81 Pb 82 Bi 83 Po 84 At 85 Rd 86
7 Fr 87 Ra 88 Ac 89
Rare Earths
Lanthanides (6) La 57 Ce 58 Pr 59 Nd 60 Pm 61 Sm 62 Eu 63 Gd 64 Tb 65 Dy 66 Ho 67 Er 68 Tm 69 Yb 70 Lu 71
Actinides (7) Ac 89 Th 90 Pa 91 U 92 Np Pu


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Question 1



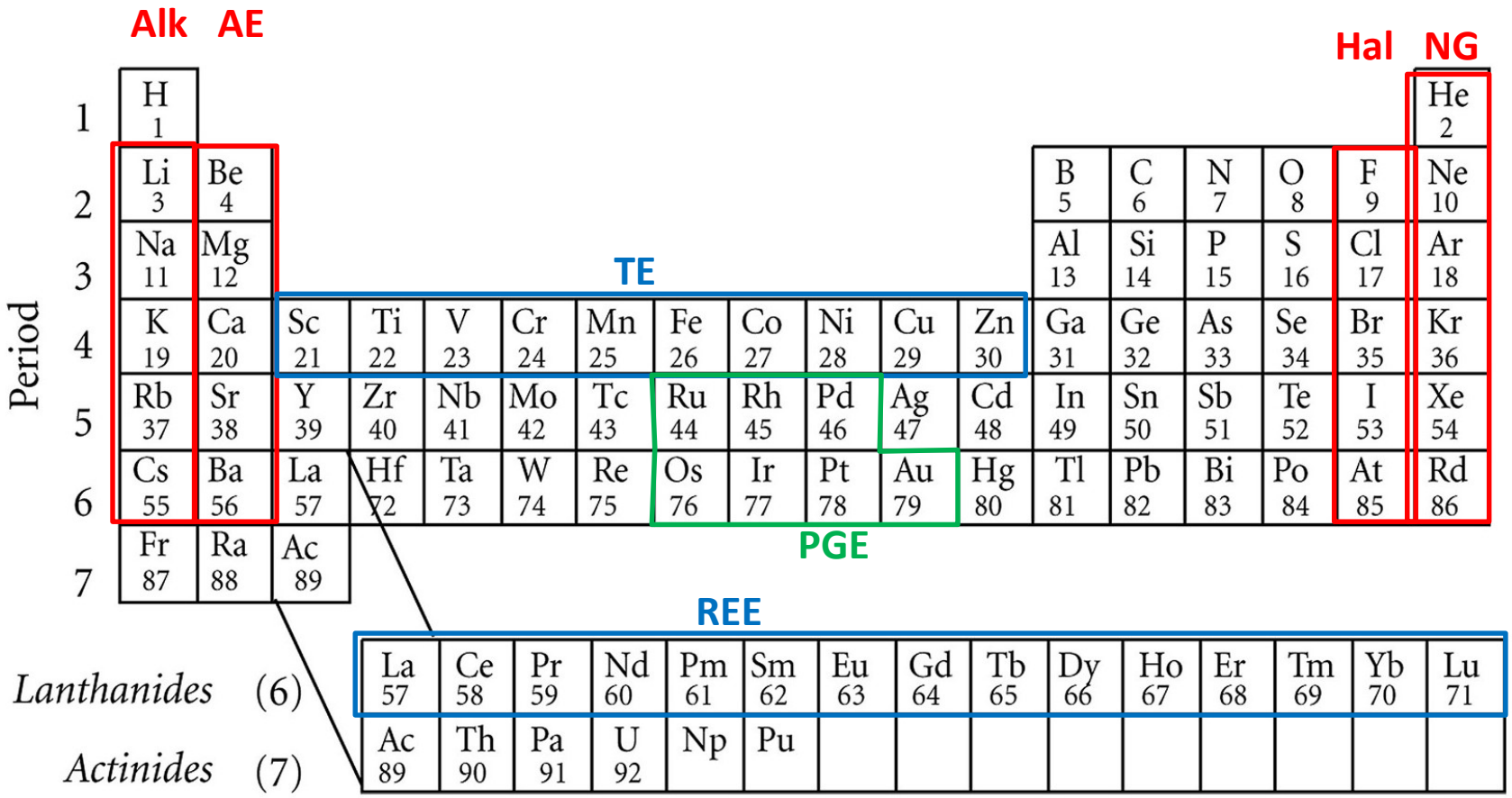
Identify:

Groups: Alkali, Alkaline Earth, Halogens, Noble Gases

Periods: Transition Elements, Rare Earth Elements

Trace elements: Platinum Group Elements, REEs

Question 1



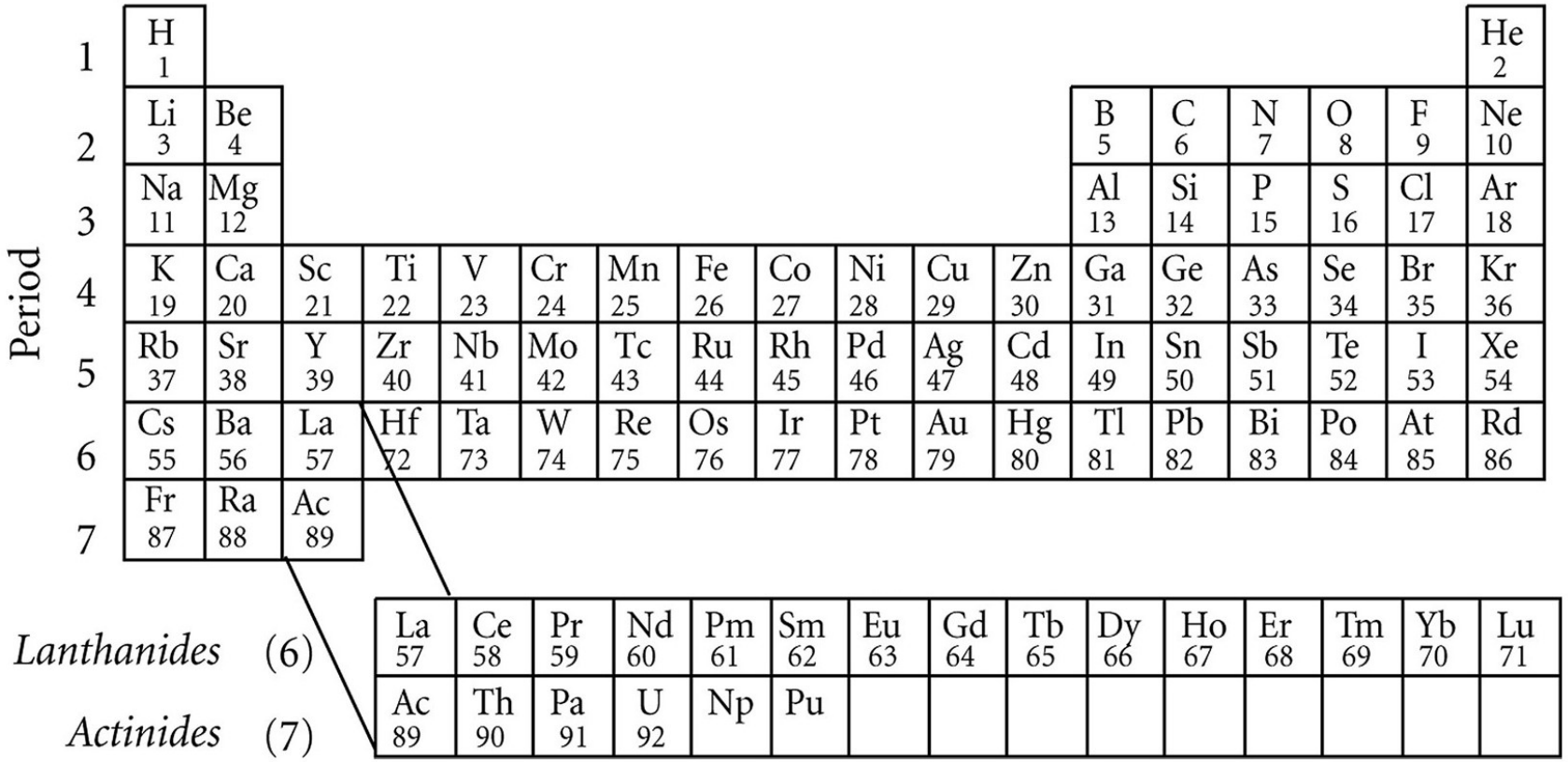
Identify:

Groups: Alkali, Alkaline Earth, Halogens, Noble Gases — **Alk, AE, Hal, NG**

Periods: Transition Elements, Rare Earth Elements — **TE, REE**

Trace elements: Platinum Group Elements, REEs — **PGE**

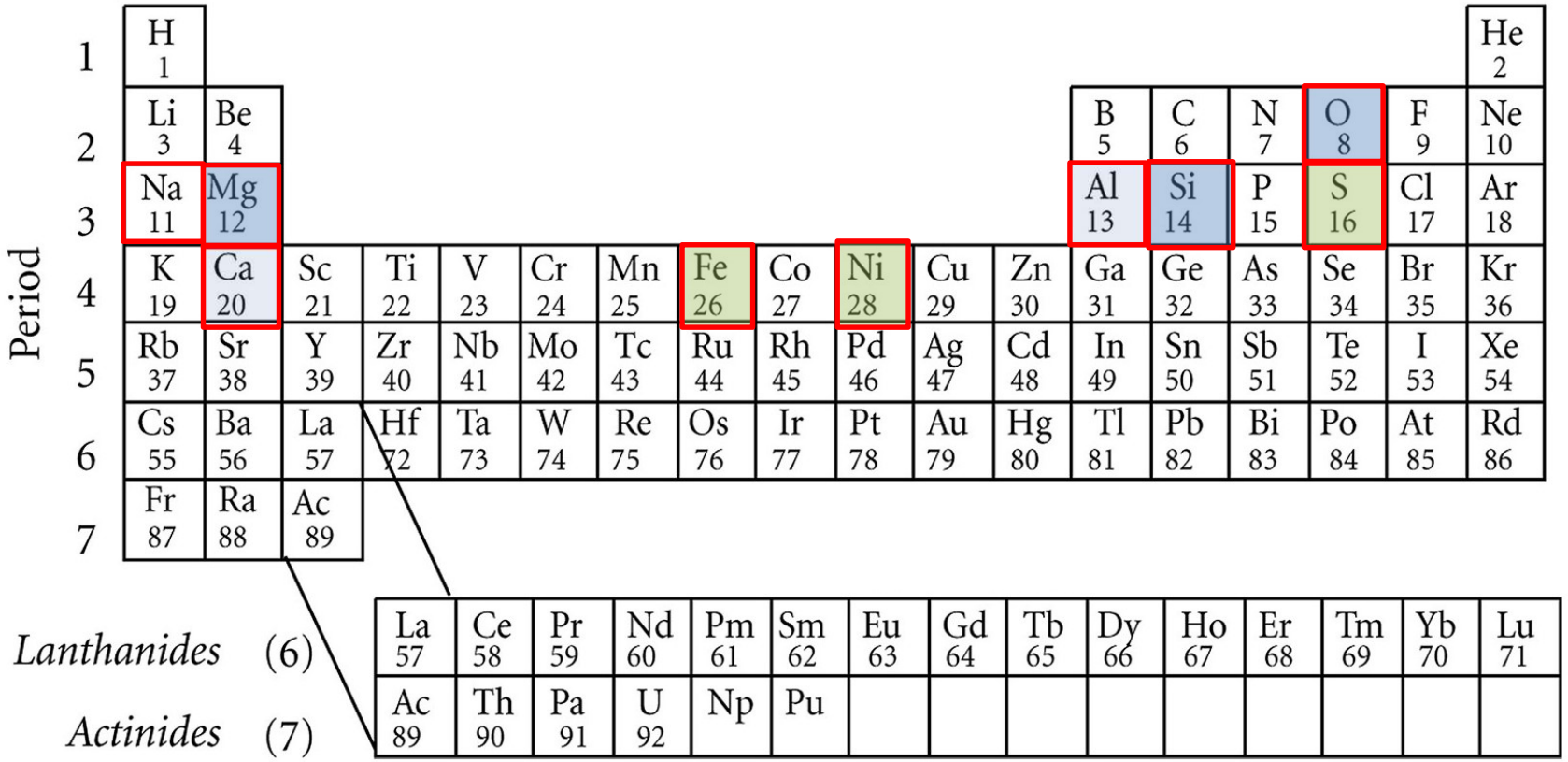
Question 2



Identify:

Most abundant elements in bulk Earth, mantle, core

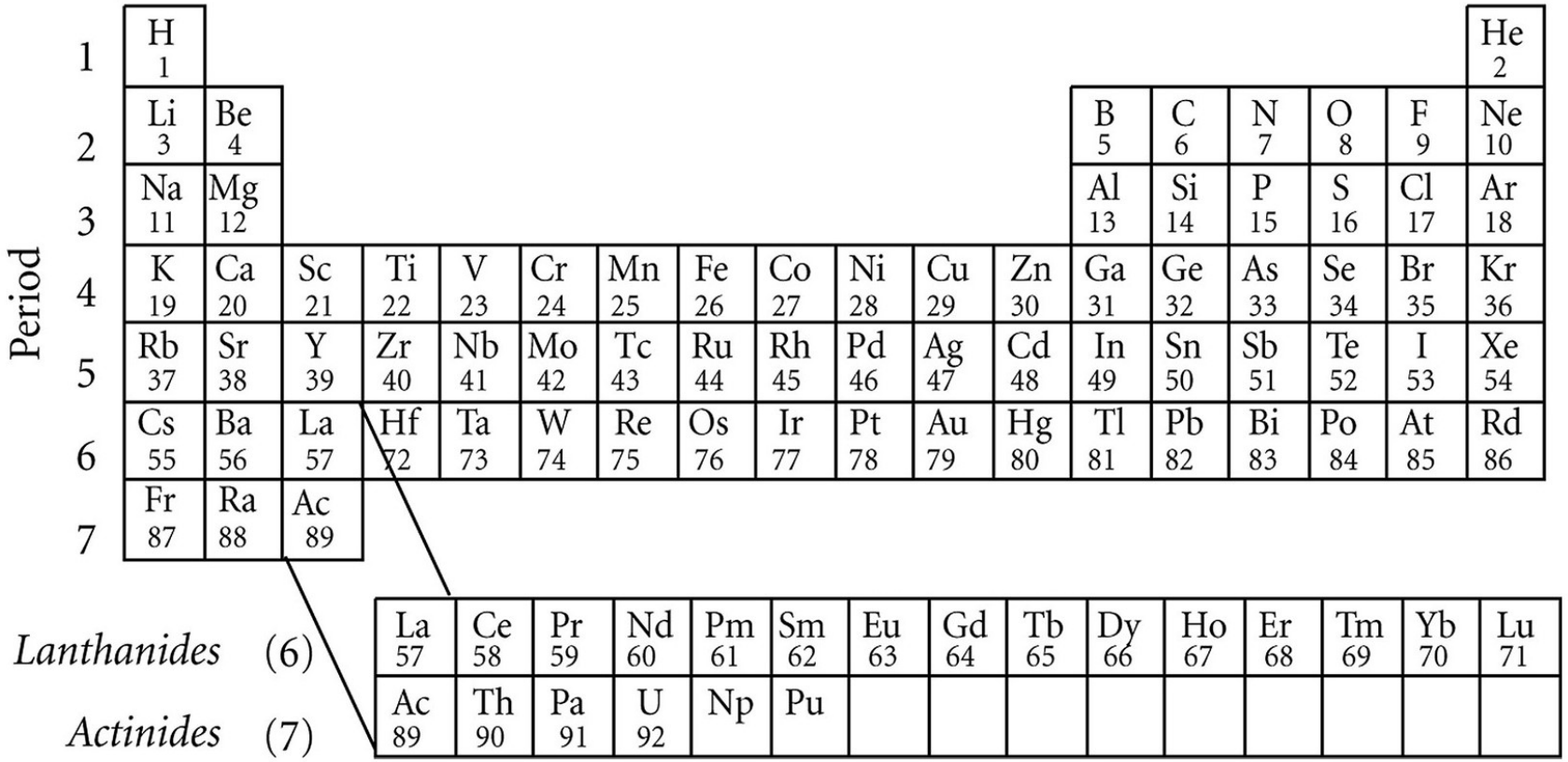
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<u>Identify:</u>

Most abundant elements in **bulk Earth**, **mantle**, **core**

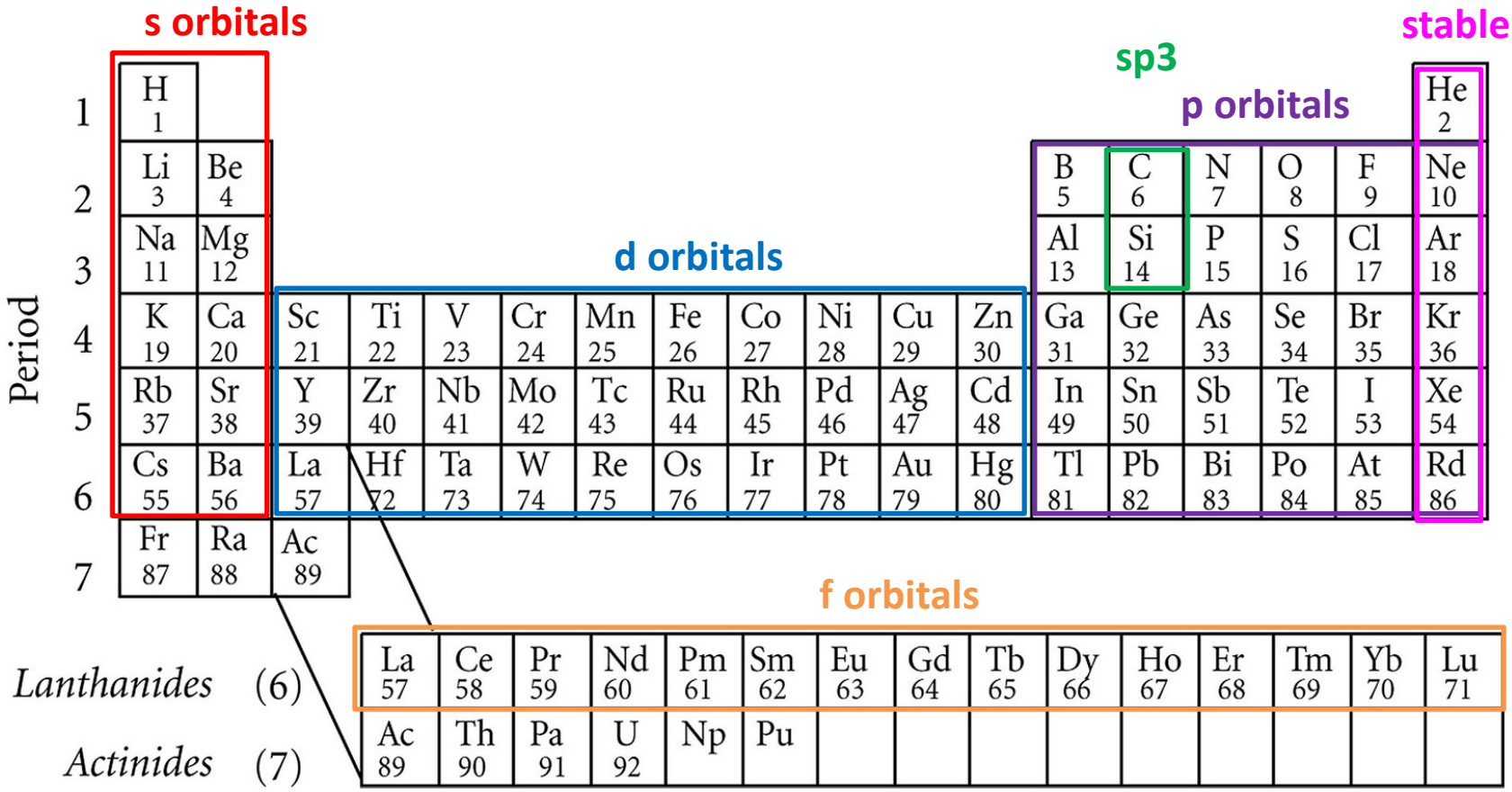
Question 3



Identify:

Orbitals: Elements that are filling s, p, d and f orbitals;
elements that form tetrahedral sp3 orbitals;
elements with filled electron shells

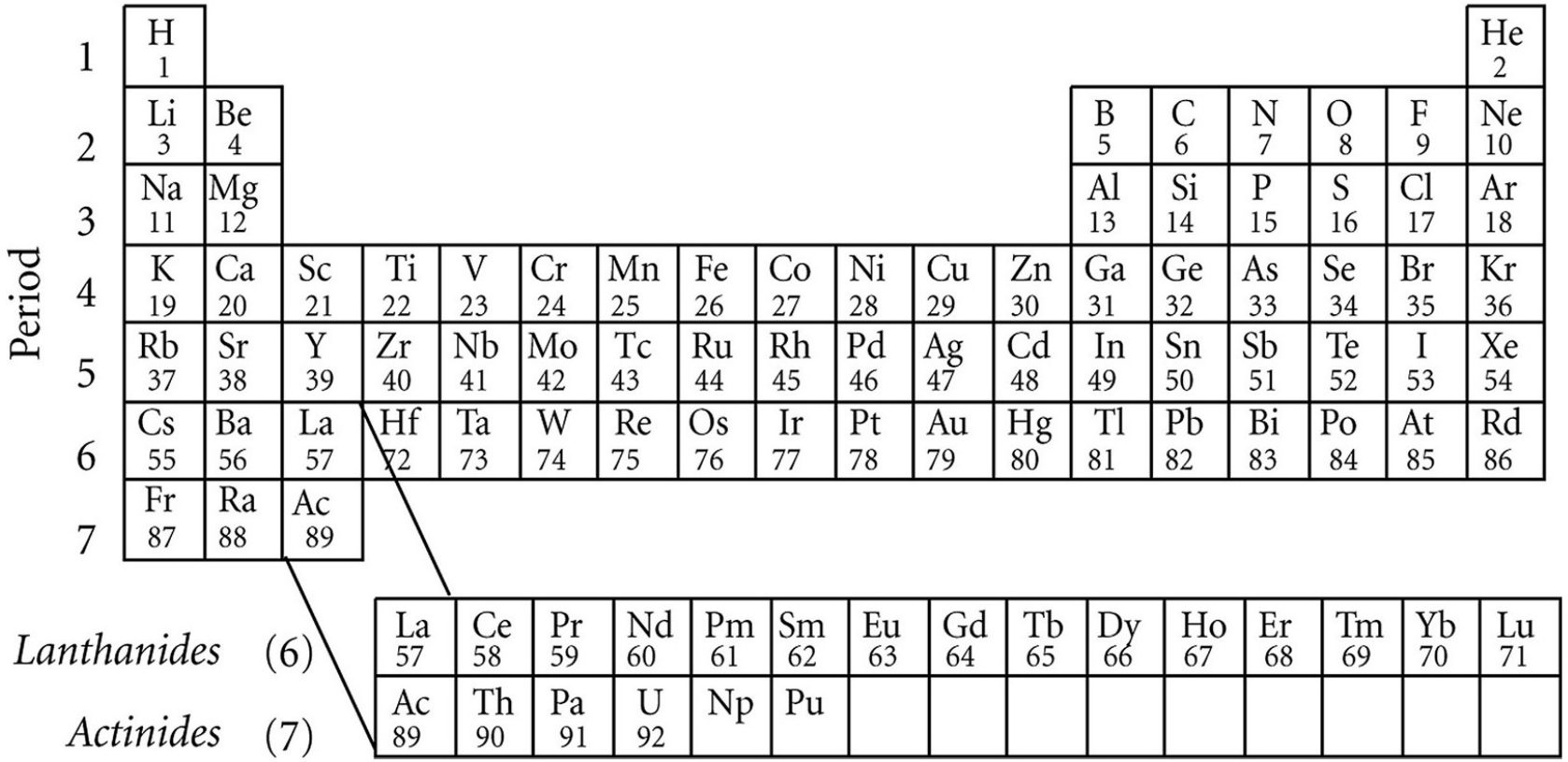
Question 3



Identify:

Orbitals: Elements that are filling **s**, **p**, **d** and **f** orbitals;
elements that form tetrahedral **sp3** orbitals;
elements with **stable electron configuration**

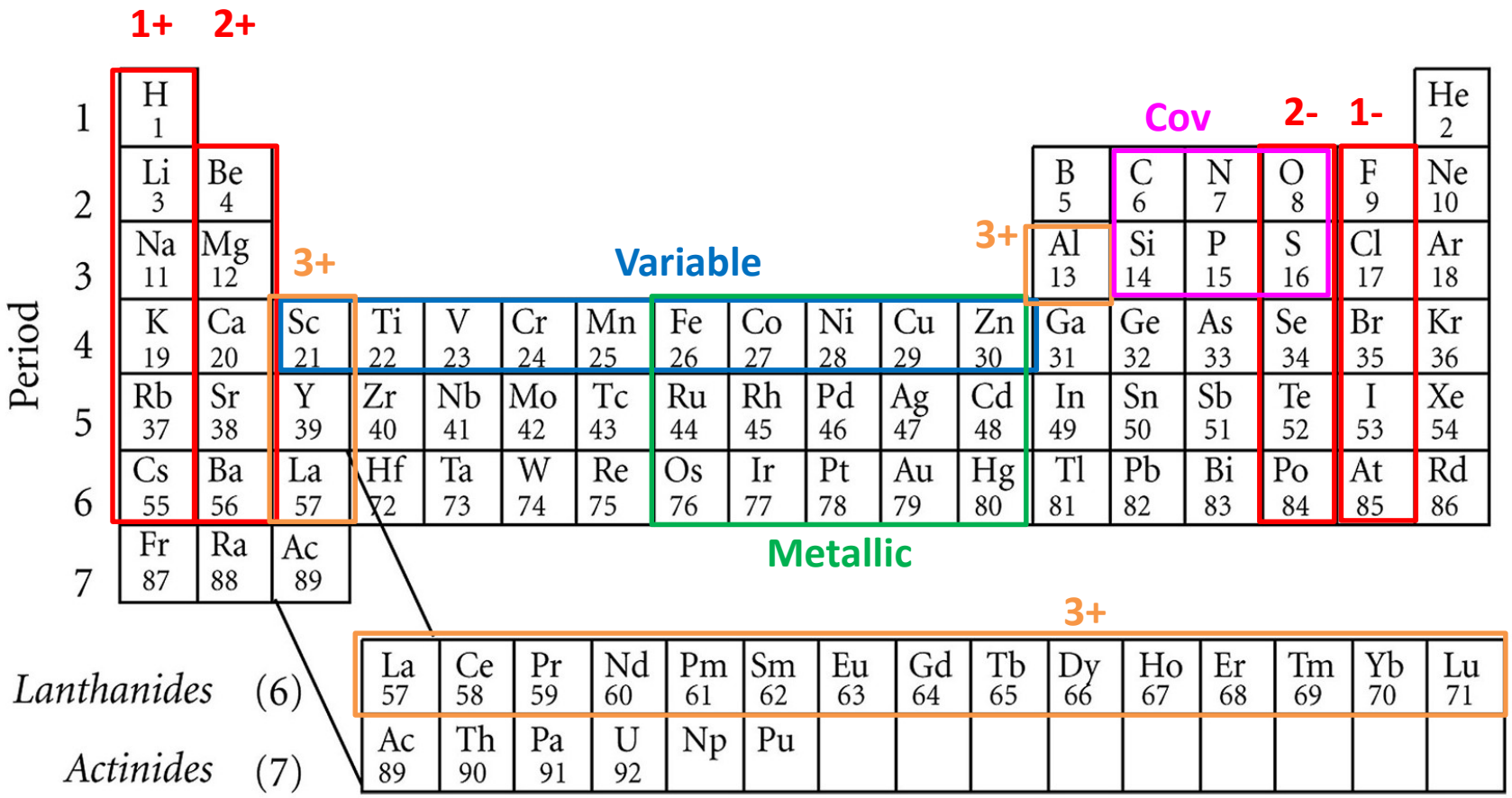
Question 4



Identify:

Bonding: Common 1+ and 2+ cations; 1- and 2- anions; elements that form 3+ cations;
elements that form covalent bonds – and hence complex ions;
elements that form metallic bonds;
elements that can have variable oxidation states (i.e. variable charge on ions)

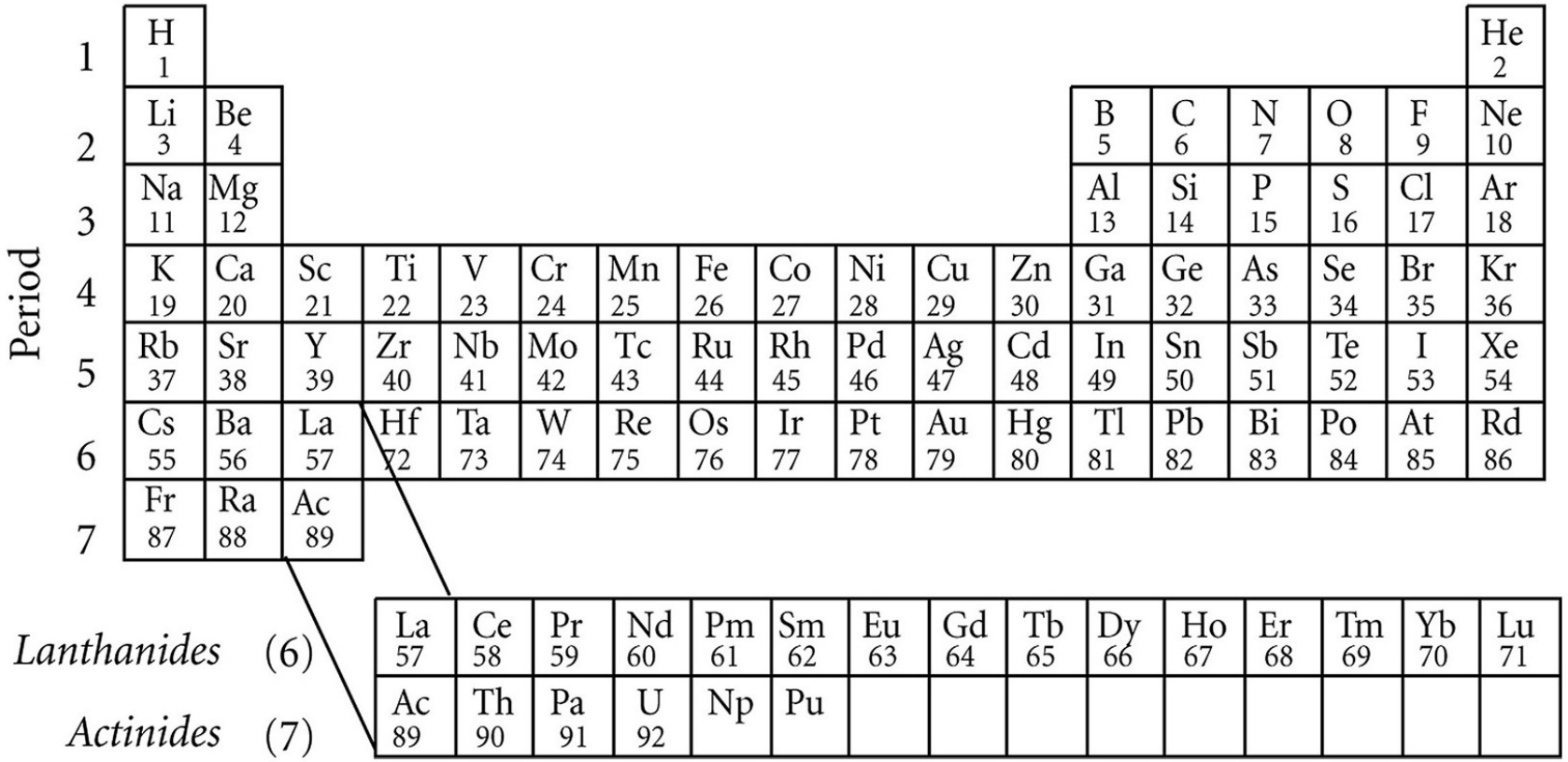
Question 4



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Bonding: Common **1+ and 2+ cations; 1- and 2- anions**; elements that form **3+ cations**;
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elements that form **metallic bonds**;
elements that can have **variable oxidation states** (i.e. variable charge on ions)

Question 5



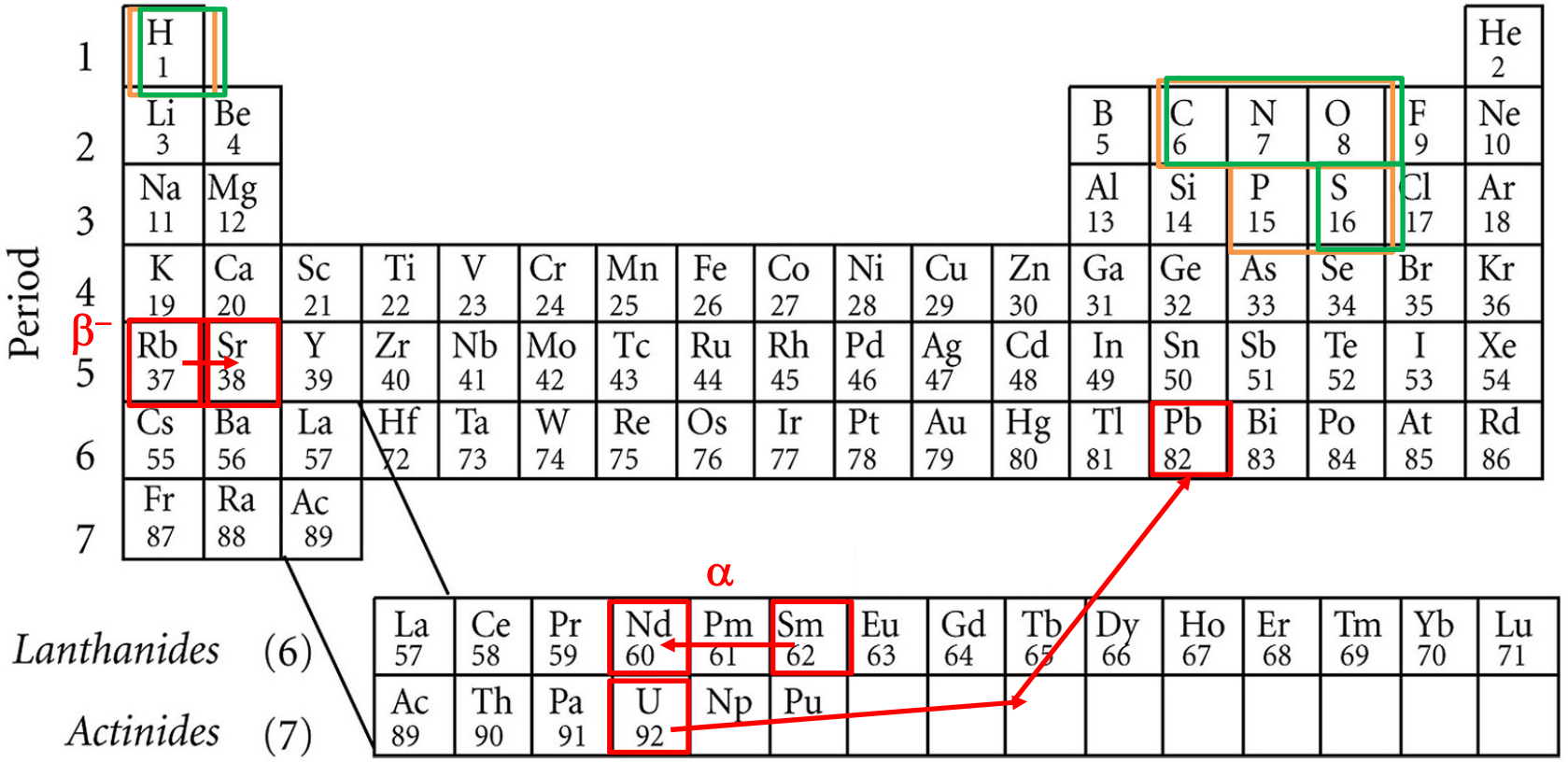
<u>Identify:</u>

Most abundant elements in **biogenic material**

Elements important for **stable isotope studies**

Elements useful for **geochronology (radioactive isotopes)**

Question 5



Identify:

Most abundant elements in **biogenic material**

Elements important for **stable isotope studies**

Elements useful for **geochronology (radioactive isotopes)**